A counterexample to positive shifted Laguerre–Pólya preservation

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Task 1. Resolve the Laguerre-operator preservation conjecture. Wagner's Conjecture 4.3 asserts that $L_j(\phi) \in \mathcal{SL}\text{-}\mathcal{P}^+$ for every $\phi \in \mathcal{SL}\text{-}\mathcal{P}^+$ and every positive integer j [1]. The operators are defined by

$$|\phi(x+iy)|^2 = \sum_{j\geq 0} L_j(\phi)(x) y^{2j}, \qquad L_1(\phi) = (\phi')^2 - \phi\phi''.$$

We disprove this assertion already for $j = 1$.

Recall the defining conditions in [1, p. 3]. Membership in the shifted class of type I requires real polynomials p_N, of degrees d_N, converging locally uniformly to ϕ, such that for each fixed d, the polynomial $p_N^{(d_N-d)}$ has all its roots real and of the same sign for every sufficiently large N. Membership in $\mathcal{SL}\text{-}\mathcal{P}^+$ additionally requires the exponential Taylor coefficients $\gamma_n = \phi^{(n)}(0)$ to be eventually nonnegative.

Theorem 1. *For any $c > 0$ and $a > 0$, the function $\phi(x) = c + e^{ax}$ belongs to $\mathcal{SL}\text{-}\mathcal{P}^+$, whereas $L_j(\phi) \notin \mathcal{SL}\text{-}\mathcal{P}^+$ for every odd positive integer j. In particular, $\phi(x) = 1 + e^x$ is a counterexample to Conjecture 4.3.*

Proof. Set $p_N(x) = c + (1 + ax/N)^N$. These real polynomials have degree N and converge to $c + e^{ax}$ uniformly on each compact subset of $\mathbb{C}$. For fixed $d \geq 1$ and $N > d$, differentiation annihilates the constant term and gives

$$p_N^{(N-d)}(x) = \frac{N!}{d!}\left(\frac{a}{N}\right)^{N-d}\left(1 + \frac{ax}{N}\right)^d. \tag{1}$$

All d roots are $-N/a < 0$. Thus the defining approximation condition holds, with threshold $N = d+1$. Moreover,

$$\gamma_0 = c + 1 > 0, \qquad \gamma_n = a^n > 0 \quad (n \geq 1),$$

so $\phi \in \mathcal{SL}\text{-}\mathcal{P}^+$.

The coefficient criterion in [1, Theorem 1.1] gives a separate check of membership: for every $m \geq 1$ and $d \geq 1$, the shifted Jensen polynomial is

$$J_\gamma^{d,m}(X) = \sum_{k=0}^{d} \binom{d}{k} \gamma_{m+k} X^k = a^m (1 + aX)^d, \tag{2}$$

whose roots are all negative.

For real x, y, direct multiplication gives

$$|c + e^{a(x+iy)}|^2 = c^2 + e^{2ax} + 2ce^{ax}\cos(ay).$$

Consequently

$$L_j(\phi)(x) = \frac{2c(-1)^j a^{2j}}{(2j)!} e^{ax} \qquad (j \geq 1). \tag{3}$$

For odd j, every exponential Taylor coefficient of this function equals $-2ca^{2j+n}/(2j)! < 0$. Eventual nonnegativity therefore fails. In particular, $L_1(1 + e^x) = -e^x \notin \mathcal{SL}\text{-}\mathcal{P}^+$. □

Scope. The counterexample has strictly positive Taylor coefficients at every index and satisfies both the direct approximation definition and the shifted Jensen criterion. Repeated negative Jensen roots are allowed in the source. The conclusion concerns the positive shifted class; the negative exponential remains in the unsigned shifted class.

Task 2. Analyze polynomial perturbations. For $a > 0$ and real polynomials P, characterize when $L_j(e^{ax} + P(x))$ belongs to $\mathcal{SL}$-$\mathcal{P}^+$, and identify which features of P control eventual coefficient signs.

Task 3. Investigate preservation after differentiation. Determine sufficient conditions on $\phi \in \mathcal{SL}$-$\mathcal{P}^+$ under which, for each j, some N satisfies $L_j(\phi^{(n)}) \in \mathcal{SL}$-$\mathcal{P}^+$ for every $n \geq N$.

Task 4. Test strict shifted hyperbolicity. Determine whether counterexamples persist when every fixed-degree shifted Jensen polynomial eventually has distinct negative roots.

Task 5. Examine the even operators. Decide whether the operators L_{2j}, which preserve positivity on the family in (3), preserve $\mathcal{SL}$-$\mathcal{P}^+$ in general, or construct counterexamples to that restricted assertion.

References

[1] I. Wagner, *On a new class of Laguerre–Pólya type functions with applications in number theory*, arXiv:2108.01827v2 (2022), pp. 3–4 and Conjecture 4.3 on p. 11. Published in *Pacific Journal of Mathematics* **320** (2022), 177–192, doi:10.2140/pjm.2022.320.177.